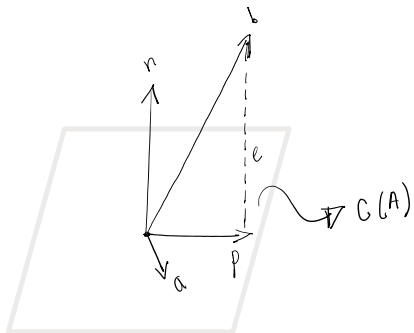


Least Square Approximations

previously, $P = A(A^T A)^{-1} A^T$ || projection vector, $p = Pb = A(A^T A)^{-1} A^T b$.

↳ Projection Matrix onto a subspace.

↳ if b is in column space $Pb = b$.
if b is $\perp$ to column space $Pb = 0$.



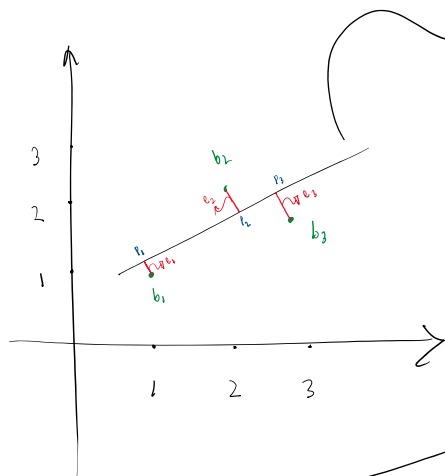
$$\begin{aligned} b &= p + e \\ \Rightarrow e &= b - p \\ \Rightarrow e &= b - Pb \\ \Rightarrow e &= (I - P)b \end{aligned}$$

↳ $Pn = 0$

↳ left nullspace $\sim N(A^T)$

$Ax = b \sim$ previously we said that if this got no solution then we can solve it using, $A^T A \hat{x} = A^T b$.

let's take a look at a real life example:



$y = C + Dt$; Here, given three points, we can try drawing a line that goes through all three points. But it is not possible since it does not satisfy $y = C + Dt$.

$$\begin{cases} C + D = 1 \\ C + 2D = 2 \\ C + 3D = 2 \end{cases}$$

These set of equations does not have a solution. But they do have a best possible solution.

(. from this scenario,

So from this scenario,

$$Ax = b$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \sim$$

So what would best possible solution mean?
Assuming we find a line, rather the best possible line that is closest to all three points. And each of these points will make an error with respect to the line. If the errors are e_1, e_2 and e_3 we have to minimize them as much as possible.

$$\text{Minimize, } \|Ax - b\|^2 = \|e\|^2 \\ = e_1^2 + e_2^2 + e_3^2$$

$$\text{Find } \hat{x} = \begin{bmatrix} \hat{C} \\ \hat{D} \end{bmatrix}, p$$

(Actual C and D does not exist. $\hat{C}$ and $\hat{D}$ are the best possible answer to C and D)

$$\text{Now, } A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\hookrightarrow A^T A x = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

$$\hookrightarrow 3C + 6D = 5$$

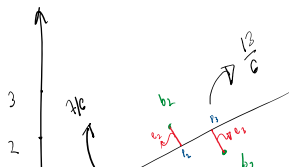
$$6C + 14D = 11$$

$$\hookrightarrow C = \frac{2}{3}, D = \frac{1}{2}$$

From the errors we are trying to minimize,

$$e_1^2 + e_2^2 + e_3^2 \\ = (C+D-1)^2 + (C+2D-2)^2 + (C+3D-3)^2$$

So, the best possible line: $y = \frac{2}{3} + \frac{1}{2}t$

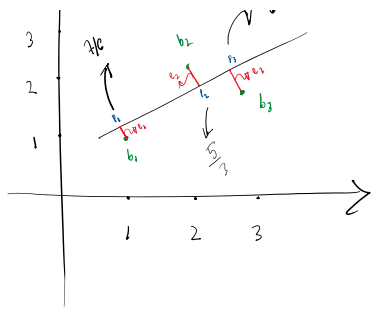


$$\text{When, } t=1, y = \frac{7}{6} \\ t=2, y = \frac{5}{3}$$

$$b = p + e \\ e = b - p$$

$$e = 1 - \frac{7}{6} = -\frac{1}{6}$$

$$e = 2 - \frac{5}{3} = \frac{1}{3}$$

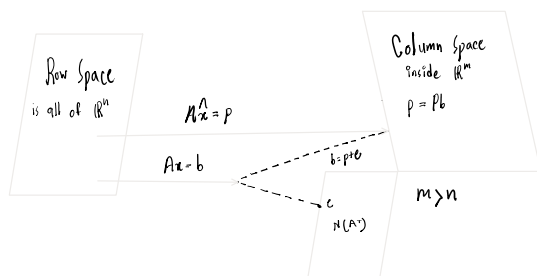


When $x=0$, $y=0$

$t=2, y = \frac{5}{3} \quad e = 2 - \frac{5}{3} = \frac{1}{3}$

$t=3, y = \frac{13}{6} \quad e = 2 - \frac{13}{6} = -\frac{1}{6}$

Big Picture for Least Square Approximation



Here the equation $Ax=b$ is not always solvable because b may not be in the column space of A .

↳ This happens when A is an $m \times n$ matrix with $m > n$, meaning the system is overdetermined.

(Meaning more equations than unknowns. We just tried solving an example)

Since b is not in the column space, we decompose it as: $b = p + e$.

↳ $p = Pb$ is the projection of b onto the column space of A .

↳ $e = b - p$ is the error vector, which is perpendicular to the column space.

↳ Instead of solving $Ax=b$, we solve $A\hat{x}=p$ which finds $\hat{x}$ such that $A\hat{x}$ is as close as possible to b .

↳ Least Square solution, $\hat{x}$ is chosen to minimize the error $e = b - A\hat{x}$.

Mathematically, the best $\hat{x}$ satisfies the normal equations: $A^T A \hat{x} = A^T b$.

This ensures that the error e is orthogonal to the column space: $A^T e = 0$.

⊗ $\hat{x}$ is the best approximate solution to $Ax=b$, and $A\hat{x}$ is the closest

point to b in the column space.

Fitting a Straight Line

↳ Fitting a straight line to a set of data points is the clearest application of least squares because it directly illustrates the fundamental idea of minimizing errors in an overdetermined system.

Let's say, we are given m data points (t_i, b_i) , where:

$$b_i \approx C + Dt_i$$

t_i = the input (independent variable)

b_i = the observed output (dependent variable)

C = intercept, D = slope.

Now, the equations:

$$C + Dt_1 = b_1$$

$$C + Dt_2 = b_2$$

$\vdots$

$$C + Dt_m = b_m$$

Since there are m equations but only two unknowns (C and D), this system is overdetermined, meaning an exact solution generally does not exist.

Now, how to solve this? Let us try to generalize it from example 1.

Step 1: Set up the problem with matrices.

$$C + Dt_1 = b_1$$

$$C + Dt_2 = b_2$$

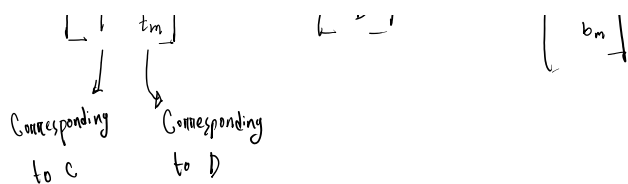
$\vdots$

$$C + Dt_m = b_m$$

Can be written as: $Ax = b$

$$A = \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}, \quad x = \begin{bmatrix} C \\ D \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$\downarrow \quad \downarrow$



Step 2: Compute $A^T A$ and $A^T b$

Using the least square method, we solve the normal equation:

$$A^T A x = A^T b.$$

Now, we compute $A^T A$:

$$A^T A = \begin{bmatrix} 1 & 1 & \dots & 1 \\ t_1 & t_2 & \dots & t_m \end{bmatrix} \begin{bmatrix} 1 \\ t_1 \\ \vdots \\ t_m \end{bmatrix}$$

$$\hookrightarrow A^T A = \begin{bmatrix} m & \sum t_i \\ \sum t_i & \sum t_i^2 \end{bmatrix}$$

Now we compute $A^T b$:

$$A^T b = \begin{bmatrix} 1 & 1 & \dots & 1 \\ t_1 & t_2 & \dots & t_m \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$\hookrightarrow A^T b = \begin{bmatrix} \sum b_i \\ \sum t_i b_i \end{bmatrix}$$

Step 3: Solving for $\hat{x} = (C, D)$

The final equation to solve for C and D is:

$$\begin{bmatrix} n & \sum t_i \\ \sum t_i & \sum t_i^2 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} \sum b_i \\ \sum t_i b_i \end{bmatrix}$$

$$\hookrightarrow \hat{x} = (A^T A)^{-1} A^T b.$$

Example 1: We have three data points:

$$t = (-2, 0, 2)$$

$$b = (1, 2, 4)$$

Assuming the data follows a linear model: $b_i \approx C + Dt_i$

$$\text{Now, } Ax = b$$

$$\hookrightarrow \begin{bmatrix} 1 & -2 \\ 1 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

$$\text{Now, } A^T A = \begin{bmatrix} 1 & 1 & 1 \\ -2 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 8 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 \\ -2 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$$

$$\hookrightarrow A^T A x = A^T b$$

$$\Rightarrow \begin{bmatrix} 3 & 0 \\ 0 & 8 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$$

$$\hookrightarrow \begin{aligned} 3C &= 7 \Rightarrow C = \frac{7}{3} \\ 8D &= 6 \Rightarrow D = \frac{3}{4} \end{aligned}$$

$$\therefore b = \frac{7}{3} + \frac{3}{4}t \Rightarrow \text{Best fit line.}$$

Example 2:

A botanist is studying the relationship between the amount of fertilizer applied (in grams) to a specific type of tomato plant and the resulting tomato yield (measured in kilograms). They conduct an experiment where they apply varying amounts of fertilizer to several plants and record the corresponding yield for each plant. Their findings can be seen in the following table:

Fertilizer Applied	Tomato Yield
0	1
1	3
2	4
3	4

Predict what the yield will be if the botanist applies 7 grams of fertilizer?

Solⁿ:

From the question:

fertilizer applied, $t = (0, 1, 2, 3)$

tomato yield, $b = (1, 3, 4, 4)$

Here, assuming the data follows a linear model: $b_i \approx c + dt_i$

$$\text{So, } Ax = b$$

$$\hookrightarrow A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad x = \begin{bmatrix} c \\ d \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 1 \\ 3 \\ 4 \\ 4 \end{bmatrix}$$

$$\text{Now, } A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 12 \\ 23 \end{bmatrix}$$

$$\text{So } A^T A x = A^T b$$

$$\hookrightarrow \begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 12 \\ 23 \end{bmatrix}$$

$$\begin{cases} 4C + 6D = 12 \\ 6C + 14D = 23 \end{cases} \Rightarrow \begin{cases} C = 3/2 \\ D = 1 \end{cases}$$

∴ Best fit line: $b = 3/2 + t$

if we apply 7 gm fertilizer: $b = 3/2 + 7 = 8.5 \text{ gm}$

Example 3:

Professor X, instructing a collegiate course, observed a post-examination trend: students reporting a higher duration of video game engagement demonstrated superior test performance. Seeking to investigate this potential correlation further, Professor X initiated a data collection effort to expand the scope of his preliminary observations. His findings can be seen in the following table:

Minutes of Games Played	Test Marks
6.1	0
7.6	2
8.7	4
10.4	6

Following this trend, if you play for 16 minutes, what marks should you get?

Solⁿ:

From the question,

minutes of games played, $t = (6.1, 7.6, 8.7, 10.4)$

Test marks, $b = (0, 2, 4, 6)$

Assuming the data follows a linear model, $b_i \approx C + Dt_i$

Now, $Ax = b$

Here,

$$A = \begin{bmatrix} 1 & 6.7 \\ 1 & 7.6 \\ 1 & 8.7 \\ 1 & 10.4 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 2 \\ 4 \\ 6 \end{bmatrix}$$

Now,

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 6.7 & 7.6 & 8.7 & 10.4 \end{bmatrix} \begin{bmatrix} 1 & 6.7 \\ 1 & 7.6 \\ 1 & 8.7 \\ 1 & 10.4 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 33.4 \\ 33.4 & 286.5 \end{bmatrix}$$

Now,

$$A^T b = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 6.7 & 7.6 & 8.7 & 10.4 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 4 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} 12 \\ 112.4 \end{bmatrix}$$

$$\therefore A^T A x = A^T b$$

$$\hookrightarrow \begin{bmatrix} 4 & 33.4 \\ 33.4 & 286.5 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 12 \\ 112.4 \end{bmatrix}$$

$$\left. \begin{aligned} 4C + 33.4D &= 12 \\ 33.4C + 285.5D &= 112.4 \end{aligned} \right\}$$

$$\hookrightarrow C = - \frac{7904}{761} = -10.4$$

$$D = \frac{1220}{761} = 1.6$$

$$\therefore \text{Best fit line: } b = -10.4 + 1.6t$$

$$\text{playing for 10 minutes, } b = -10.4 + (16 \times 10) = 15.2$$